

Technology Plus

Unit 3: Hardware

2.4.1 – Install/Configure Peripherals

Lesson Overview:

In this lesson, students will learn about peripheral devices and the basics of installing and configuring the devices.

Students will:

- Identify common types of peripheral devices.
- Identify and compare the various installation types of peripheral devices.
- Describe the basic steps of installing and configuring peripheral devices.

Guiding Question: What are peripheral devices, and how are they installed and configured for the user's benefit?

Suggested Grade Levels: 8-12

Technology Needed: Yes

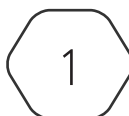
Approximate Time Required: 50-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.4 – Given a scenario, install and configure common peripheral devices.
 - Devices
 - Printer
 - Scanner
 - Keyboard
 - Mouse
 - Web camera
 - External drive
 - Speakers/headset
 - Display
 - Smart TV
 - Projector
 - Monitor
 - Uninterruptable power supply (UPS)
 - Installation types
 - Plug-n-play vs. driver installation
 - Other required steps
 - IP-based peripherals
 - Web-based configuration steps

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Lesson Background

Background Information:

Modern computing systems rely on a variety of external devices to extend functionality and enhance user experience. These devices are commonly referred to as peripherals, and include printers, keyboards, webcams, scanners, and input/output software. While many peripherals are designed to be plug-and-play, proper installation and configuration are essential for optimal performance, compatibility, and security.

Students will explore installing and configuring an IP-based peripheral device using web-based configuration tools. This will help students gain practical skills in device setup and troubleshooting that are applicable in the real-world.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.4.1 – Install Configure Peripherals • Lab Slides 2.4.1 – IP-Based Printer Configuration
Materials per Student	• Student Notes 2.4.1 – Install Configure Peripherals • Student Handout 2.4.1 – Install Configure Peripherals • Assessment 2.4.1 – Install Configure Peripherals (optional)

Lesson Background

Cyber Terms:

Term	Definition
peripherals	external devices that connect to a computer to add new features, improve performance, or provide additional input or output options
plug-and-play (PnP) device	devices automatically recognized and configured by an OS without needing additional drivers or manual intervention
driver installation device	devices that are not automatically recognized by the OS and require specific drivers to be installed for full functionality
IP-based peripherals	peripheral devices that connect to a user's network rather than directly to their computer

2.4.1 – Install/Configure Peripherals

Guiding Question: What are peripheral devices, and how are they installed and configured for the user's benefit?

Lesson Launch (10 minutes)

- Show students the various peripheral devices on slide 3.
- **Ask:** What do you think life would be like without these devices?
- Allow thinking time before students share.
- **Answer:** Students should recognize that a computer can still function without peripherals, but peripherals make technology easier, faster, and more enjoyable to use. They expand what users can do, like print, play games, or interact more comfortably, but the core computer can still operate without them.
- **Say:** These devices, known as peripherals, play an important role in enhancing the functionality and usability of computers and other devices.
- Share the definition of **peripherals**, what installation/configuration is, and basic installation/configuration steps for any device.
 - Installation is the process of setting up software or hardware so it can start working on a system.
 - Configuration is the process of customizing or adjusting settings after installation to ensure the system or device works the way a user wants.
 - Installation is like assembling a new desk, configuration is adjusting the chair height, drawer layout, or lamp position to make it fit your needs or wants.
 - Basic installation/configuration steps are for connecting peripherals to a computer.

Peripheral Devices: Enhance System Functionality (20-30 minutes)

- Distribute Student Handout 2.4.1 – Install/Configure Peripherals and encourage students to fill in the answers during the lesson.
- Discuss **printers**, their definition, common types, installation steps, and configuration options.
 - Inkjet printers spray tiny drops of liquid ink directly onto paper to print text and images on paper.
 - Laser printers use laser beams, electric charges, and powdered toner to print text and images on paper.
 - 3D printers create three dimensional objects using resin or filament.
- Discuss **scanners**, their definition, installation steps, and configuration options.
 - Scanners can be purchased as an individual device or can come integrated with a printer.
- Discuss **keyboards**, their definition, installation steps, and configuration options.

- Each key on a keyboard has a specific code – when a key is pressed, the keyboard sends the code to the operating system and displays the associated letter, number, or function on the screen.
- Keyboards are typically plug-and-play (PnP) – no driver installation necessary.
- Discuss **mice**, their definition, installation steps, and configuration options.
 - Some devices, like laptops, have touchpads as their pointing devices. A physical mouse can still be connected to a laptop via wired or wireless connections.
 - Mice are typically plug-and-play; some can have software installation for enabling advanced features.
- Discuss **web cameras**, their definition, installation steps, and configuration options.
 - Web cameras can be built into laptops or plugged in via USB connections.
 - Web cameras can also be Bluetooth/wireless.
- Discuss **external drives**, their definition, installation steps, and configuration options.
 - Some external drives require a power source to be connected.
 - Most external drives are plug-and-play.
- Discuss **speakers/headsets**, their definition, installation steps, and configuration options.
 - Speakers and headsets can connect via audio jack, USB, and Bluetooth.
- Discuss **displays**, their definition, installation steps, and configuration options.
 - Graphics cards send visual data as signals and the display devices send the signal through wired connections.
 - Display devices connect to computers via HDMI, VGA, DisplayPort, or USB connections.
 - Display devices also consist of smart TVs, projectors, and monitors.
- Discuss **uninterruptable power supply (UPS)**, its definition, installation steps, and configuration options.
 - UPS gives a system time to shut down safely or keep running briefly, it also has filters to stop damage to electronics from electrical spikes and surges.
 - Introduce different installation types and other required steps.
 - All the devices you learned about today are external - because peripheral devices are external devices.
 - Common **plug-and-play (PnP)** devices include modern keyboards, mice, and printers.
 - Common devices that require driver installation are high-end printers, scanners, and specialized gaming peripherals.
- Discuss **IP-Based Peripherals** and web-based configuration.
 - Common IP-based peripherals include network printers, IP cameras, and network-attached storage devices.
 - Not all, but most IP-based peripherals use web-based configuration.
 - Dynamic Host Configuration Protocol (DHCP) - a service that automatically gives IP addresses to

devices on a network so they can connect without manual setup.

- A static IP is manually set by the user and does not change.
- Example of testing configuration: print test page, check video feed, test user permissions.
- Discuss common connection types.

Lab: IP-Based Printer Configuration (15-25 minutes)

- This lab requires access to the CYBER.ORG Range – you will need to launch the Ubuntu Linux machine and Lab Slides 2.4.1 – IP-Based Printer Configuration
- Introduce the guiding question and objectives for the lab.
- In this lab, students will use Common UNIX Printing System (CUPS) to simulate installing and configuring an IP-based printer using web-based configuration.
- Students should complete the lab section of their student handout as they work through the lab.

Helpful Tips

- **sudo systemctl start cups** makes the CUPS webserver accessible.
- **sudo usermod -aG lpadmin \$USER** grants our machine access to admin permissions on the CUPS web server.
- Username: ubuntu Password: password
- For lab purposes students should choose the Internet Printing Protocol (http) – outside of the lab, students would choose which best describes their printer based on the device manual.
- Ideally, the connection of the device would be the <device-id>. For lab purposes students will make a fake connection/IP address for the device. In the real world, the device would already have an assigned IP, or the user could assign a static IP and then type it for the connection.
- Allow students to choose the name, description, and location of the printer.
- Allow students to choose the make and model of the printer.
- Student choice is encouraged for the purposes of this lab, but students should understand that in the real world they would make their choices based on the actual device and its manual instructions.
- Based on student choices, some of their configuration options may differ from what is on the lab slides.

Putting it All Together

- What is CUPS and what role does it play in a Linux environment?

CUPS (Common UNIX Printing System) is a print server that enables printing to local and network printers using IPP, offering a web interface for management.

- How did we access the CUPS web interface?

We opened the browser and typed `http://localhost:631`

- What configuration features did you find on the Printers page?

Modify Printer

Delete Printer
Set Allowed Users
Set As Server Default
Set Default Options
Print Test Page
Pause Printer
Reject Jobs
Move All Jobs
Cancel All Jobs

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What is a peripheral device, and why are they important to computer users?

Peripherals are external devices that expand a computer's functionality, making tasks easier, faster, or possible at all (e.g., printing, scanning, displaying, or audio).
- Why is it important to understand both installation and configuration? Can you give an example of how a device might "work" but not be fully configured?

A printer might install automatically, but without configuring paper size or default options, the user won't get the results they need. Understanding both ensures devices work correctly and securely.
- What are the main installation types for peripherals, and how do they differ?

Plug-and-play installs automatically, driver installation requires specific software, IP-based peripherals connect through a network, and web-based configuration is managed through a browser.
- Imagine your school just added a new network printer. What steps would you take to install and configure it so everyone in your class could use it?

Answers will vary, but should include connecting to the network, locating or assigning an IP address, accessing the web-based interface, installing drivers if necessary, and testing with a sample print.